

E.2.1 Ensor et al. (2013)

Ensor et al. (2013) studied the association between short-term ambient air pollution (PM_{2.5} and O₃) exposure and out-of-hospital cardiac arrest (OHCA). Ensor et al. (2013) gathered medical and population subgroup data for all ages from an Emergency Medical Services database in Houston, Texas between 2004 and 2011. Authors assessed the medical data and defined out-of-hospital cardiac arrest as emergency medical services performing chest compressions. Authors collected ambient air pollution and weather data from Texas Commission of Environmental Quality monitors and calculated hourly and daily averages for PM_{2.5} and O₃. The authors used a time-stratified case crossover analysis and conditional logistic regression.

Out-of-Hospital Cardiac Arrest

In a single-pollutant model, the coefficient and standard error are estimated from a reported excess risk of OHCA of 3.9 percent (95% CI: 0.5 -7.4) for a 6 µg/m³ increase in the averaged daily mean PM_{2.5} concentration 0- and 1-days prior to onset (Ensor et al. 2013, Table 4).

E.2.2 Gharibvand et al. (2017)

Gharibvand et al. (2017) evaluated whether positive associations exist between PM_{2.5} exposure and incidence of lung cancer in non-smokers among the Adventist Health and Smog Study-2 (AHSMOG-2), a group of health-conscious individuals of which 81% are never smokers. Authors collected ambient air pollution data (PM_{2.5} and O₃) from the US EPA Air Quality system over two years (January 2000-December 2001). Three *a priori* factors were added to the models as covariates: time spent outdoors, residence length, and moving distance during follow-up. Authors modeled the association between PM_{2.5} exposure and incidence of lung cancer using a Cox proportional hazards regression, with attained age as the time variable. The authors conducted both a single and a two-pollutant (PM_{2.5} and O₃) analyses. The study concluded that each 10 µg/m³ increase in ambient PM_{2.5} concentrations was positively associated with increased lung cancer risks within the single-pollutant and two-pollutant multivariable models with O₃.

Incidence, Lung Cancer

In a two-pollutant multivariable model with O₃ (including *a priori* covariates), the coefficient and standard error were estimated from a hazard ratio of 1.46 (95% CI: 1.13-1.89) for each 10 µg/m³ increase in mean monthly ambient PM_{2.5} concentrations (Gharibvand et al. 2016, Table 3).

E.2.3 Kioumourtzoglou et al. (2016)

Kioumourtzoglou et al. (2016) evaluated the potential impact of long-term PM_{2.5} exposure on first hospital admission for dementia, Alzheimer's, or Parkinson's diseases among Medicare beneficiaries (>= 65 years old) in 50 cities in the northeastern U.S. (Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and Washington, D.C.). Authors

retrieved medical data from the Center for Medicaid and Medicare from the years 1999-2010. The study followed enrollees as a cohort, which included annual follow-up records identifying the first hospital admissions for dementia (ICD-9 290), Alzheimer's (ICD-9 331.0), Parkinson's (ICD-9 332), and other cardiovascular comorbidities. With respect to Alzheimer's disease, the study evaluated 9,817,806 Medicare enrollees and included 266,725 cause-specific hospital admissions indicating disease onset. With respect to Parkinson's disease, the study evaluated 9,817,806 Medicare enrollees and included 119,425 cause-specific hospital admissions indicating disease onset. Annual average PM_{2.5} concentrations were estimated for each city using data from the U.S. EPA Air Quality System database. Kioumourtzoglou et al. (2016) fit a time-varying Cox proportional hazards model for each city, using the city-wide annual PM_{2.5} concentrations as the time-varying exposure of interest and a linear term for the calendar year. This eliminated the impact of PM_{2.5} variation by city and any PM_{2.5} trends within cities. The model adjusted for cardiovascular comorbidities, and incorporated a counting process extension which created an observation for each year of follow-up per person. The results were then pooled across individuals and cities.

Incidence, Alzheimer's Disease (ICD-9 331.0)

In a single-pollutant model, the coefficient and standard error were estimated from a hazard ratio of 1.15 (95% CI: 1.11-1.19) for a 1 µg/m³ increase in the average annual PM_{2.5} concentrations (Kioumourtzoglou et al. 2016, Table 1).

Incidence, Parkinson's Disease (ICD-9 332)

In a single-pollutant model, the coefficient and standard error were estimated from a hazard ratio of 1.08 (95% CI: 1.04-1.12) for a 1 µg/m³ increase in the average annual PM_{2.5} concentrations (Kioumourtzoglou et al. 2016, Table 1).

E.2.4 Kloog et al. (2012)

Kloog et al. (2012) analyzed the effects of long- and short-term PM_{2.5} exposure on hospital admissions due to strokes with a new PM_{2.5} exposure model in New England (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont) from 2000 to 2006. We use this endpoint as a surrogate for PM_{2.5}-attributable stroke incidence. Authors collected medical data from 67,678 adults aged 65 to 99 in the U.S. Medicare program database from 2000 to 2006. They defined all respiratory, cardiovascular disease, stroke, and diabetes based on emergency department visits and primary discharge diagnosis records. Authors used a hybrid exposure technique comprised of daily PM_{2.5} concentration data from aerosol optical depth (AOD) measurements and ambient air monitors from the U.S. EPA and Interagency Monitoring of Protected Visual Improvements (IMPROVE). Authors also obtained land use regressions, meteorological data (National Climatic Data Center), and socioeconomic data (U.S. Census Bureau) matched to zip codes in order to perform land use Poisson regressions.

Incidence, Stroke (ICD Codes 430-436)

In a single-pollutant model for patients over the age of 65, the coefficient and standard error were estimated from the percent change (3.49%) and 95% confidence interval (0.09-5.18%) for a 10 $\mu\text{g}/\text{m}^3$ increase in the 7-year mean $\text{PM}_{2.5}$ concentrations (Kloog et al., 2012, Table 3).

E.2.5 Peters et al. (2001)

Peters et al. (2001) studied the relationship between increased particulate air pollution and onset of heart attacks in the Boston area from 1995 to 1996. The authors used air quality data for PM_{10} , $\text{PM}_{10-2.5}$, $\text{PM}_{2.5}$, "black carbon", O_3 , CO, NO_2 , and SO_2 in a case-crossover analysis. For each subject, the case period was matched to three control periods, each 24 hours apart. In univariate analyses, the authors observed a positive association between heart attack occurrence and $\text{PM}_{2.5}$ levels hours before and days before onset. The authors estimated multivariate conditional logistic models including two-hour and twenty-four hour pollutant concentrations for each pollutant. They found significant and independent associations between heart attack occurrence and both two-hour and twenty-four hour $\text{PM}_{2.5}$ concentrations before onset. Significant associations were observed for PM_{10} as well. None of the other particle measures or gaseous pollutants was significantly associated with acute myocardial infarction for the two hour or twenty-four hour period before onset.

The patient population for this study was selected from health centers across the United States. The mean age of participants was 62 years old, with 21% of the study population under the age of 50. In order to capture the full magnitude of heart attack occurrence potentially associated with air pollution and because age was not listed as an inclusion criteria for sample selection, we apply an age range of 18 and over in the C-R function. According to the National Hospital Discharge Survey, there were no hospitalizations for heart attacks among children <15 years of age in 1999 and only 5.5% of all hospitalizations occurred in 15-44 year olds (Popovic, 2001, Table 10).

Acute Myocardial Infarction, Nonfatal

The coefficient and standard error are calculated from an odds ratio of 1.62 (95% CI 1.13-2.34) for a 20 $\mu\text{g}/\text{m}^3$ increase in twenty-four hour average $\text{PM}_{2.5}$ (Peters et al., 2001, Table 4, p. 2813).

Incidence Rate: We use the county-specific daily AMI hospitalization rate (ICD-9 code 410) for the population of individuals aged 18 years and older as the estimate for the incidence rate of nonfatal heart attack, assuming all heart attacks that are not instantly fatal will result in a hospitalization. We did not adjust for fatal AMIs in the incidence rate estimation, due to the way that the epidemiological studies are designed. Those studies consider total admissions for AMIs, which includes individuals living at the time the studies were conducted. Therefore, we use the definition of AMI that matches the definition in the epidemiological studies.

Population: Population of ages 18 and older

Adjustment: As some fraction of the admitted individuals die in the hospital, we apply a survival rate of 93% in calculating the avoided cases of AMI in order to avoid double counting (once in the calculation of AMI cases and once in the calculation of PM-related mortality).

E.2.6 Pope et al. (2006)

Pope et al. (2006) evaluated the association between short-term exposure to PM_{2.5} and acute ischemic heart disease events, including acute nonfatal myocardial infarction, all acute coronary events, and subsequent myocardial infarctions in individuals living in greater Salt Lake City, Utah. In a case-crossover study, these ischemic events were assessed in relation to a 10 µg/m³ increase in PM_{2.5}. The researchers determined that a 10 µg/m³ increase in PM_{2.5} resulted in a 4.5% increase (95% CI: 1.1-8.0) in unstable angina and myocardial infarction.

Acute Myocardial Infarction, Nonfatal

In a single-pollutant model the coefficient and standard error were estimated from the percent increase (4.81%) and 95% confidence interval (95% CI: 0.98-8.79) for a 10 µg/m³ increase in daily 24-hour mean PM_{2.5} (Pope et al., 2006, Table 3).

Incidence Rate: AMI hospital admission rate for all ages. See the incidence rate discussion under Peters et al. (2001) in Section E.2.5.

Population: Ages 18 and older

Adjustment: See the adjustment description in Section E.2.5.

E.2.7 Rosenthal et al. (2008)

Rosenthal et al. (2008) examined the effects of short-term PM_{2.5} exposure on out-of-hospital cardiac arrest incidence and whether these effects were connected to population subgroup data or presence of heart rhythm. Additionally, Rosenthal et al. (2008) compared exposure time and measurement method on the effects of short-term PM_{2.5} exposure and out-of-hospital cardiac arrest incidence. Authors obtained medical data from the Wishard Ambulance Service, a local emergency medical service in Indianapolis, Indiana, from July 2, 2002 to July 7, 2006. The study defined out-of-hospital cardiac arrest using the same criteria as Ensor et al. (2013) and Silverman et al. (2010). Authors collected daily and hourly PM_{2.5} concentrations from two City of Indianapolis monitoring sites and using two separate methods: the Federal Reference Method (FRM) for 24-hour filter samples, and a Federal Equivalence Method (FEM). The authors used a case crossover analysis with conditional logistic regressions in order to study the effects of short-term PM_{2.5} exposure on out-of-hospital cardiac arrest incidence. Rosenthal et al. (2008) found a positive but statistically insignificant association between non-dead on arrival (DOA) out-of-hospital cardiac arrest cases and ambient PM_{2.5} concentrations. Although they also noted a statistically significant